07-PHOTOELECTRIC EFFECT

PURPOSE of the EXPERIMENT

1. Explanation of the photoelectric effect by observing it experimentally.

2. Finding Planck's constant (*h*) with the help of the photoelectric effect.

3. Representation of the maximum kinetic energy of photoelectrons as a function of the frequency of light.

4. Calculation of the work function and threshold frequency of the metal at the photocathode.

GENERAL INFORMATION

The photoelectric effect was first observed by H. Hertz in 1887. While Hertz was experimenting with electromagnetic waves, he noticed that the electric arcs formed in the air gap between the cathode and the anode formed more easily when ultraviolet light was sent to the cathode. Hertz himself did not dwell on this observation. However other physicists tried to understand this effect. It soon became clear that the cause of this effect was the emission of electrons from the cathode when the frequency of light incident on the cathode was high enough. Thus, light had the effect of removing electrons from a metal surface. This phenomenon is called the "photoelectric effect". The electrons removed by light are called "photoelectrons".

The fact that light has the effect of removing electrons from a metal surface is a phenomenon that can be explained by the classical theory of electromagnetic waves. For this, it is sufficient to consider that electromagnetic waves are composed of electric and magnetic fields oscillating in directions perpendicular to each other.

The electric field of an electromagnetic wave exerts a uniform force on a charged particle, the electron. Here, $\vec{E}$ is the electric field and e is the charge of the electron. Due to the repulsion caused by this force, an electron can be removed from a metal surface. For this reason, the photoelectric phenomenon did not initially surprise physicists and was thought to be explainable by classical physics. However, more detailed experiments on the photoelectric phenomenon showed that it could not be explained by classical physics.

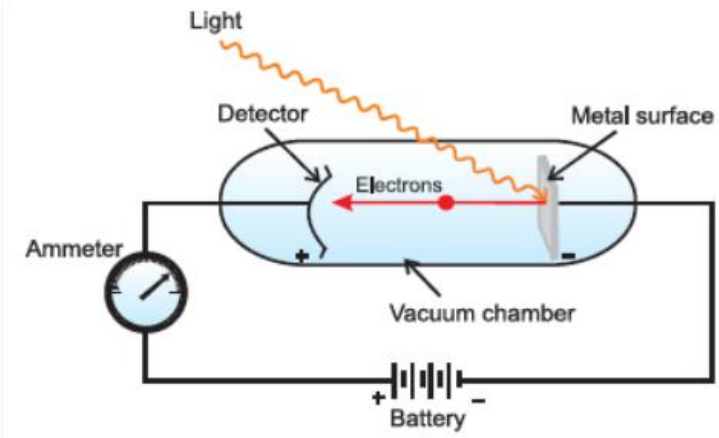


Figure 1: Photoelectric effect test tube

In 1902, P. E. A. Lenard conducted experiments to determine how the energy of photoelectrons removed from a metal plate by light depends on the intensity of the light incident on the plate. For this purpose, he illuminated a metal plate using a carbon arc lamp with adjustable light intensity.

He connected the collector plate to the cathode of a battery (Figure 1). The collector plate is thus negatively charged and a repulsion is created between the photoelectrons and the collector plate. Due to this stopping potential barrier, not all photoelectrons can easily reach the collector plate. Only those photoelectrons whose kinetic energy is large enough to overcome this stopping potential barrier can reach the collector plate. If the voltage applied by the battery is increased, no photoelectrons will reach the collector plate after a certain V value. This voltage V should be the maximum value of the kinetic energy of the photoelectrons. Thus, the photoelectrons reaching the collector plate generate a plate current which can be determined with the help of an ammeter.

$$eV_d = K_{max} \qquad \textbf{(1)}$$

According to the classical electromagnetic wave theory of light, as the intensity of light increases, the value of the electric field accelerating the electrons on the metal surface also increases. This means that the kinetic energy of the photoelectrons increases. This prediction is not consistent with the results of Lenard's experiments. Lenard repeated his experiment using light of a different color. He found that the kinetic energy of photoelectrons depends on the frequency of the light. When high-frequency light was used, the kinetic energy of the photoelectrons was also large.

The results of these experiments, which could not be explained by the classical electromagnetic theory of light, were announced by A. Einstein in 1905. Einstein predicted that the energy of light is not a continuous distribution of energy scattered along wavefronts, as predicted by the classical theory, but is concentrated in specific packets. In making this prediction, Einstein used the assumption that Planck had used to explain blackbody radiation. In 1900, to explain the nature of black body radiation, Planck proposed that the energies of the standing electromagnetic wave modes in a cavity could be analyzed as a function of their energy,

$$E_n = nh\nu \qquad \textbf{(2)}$$

Einstein assumed that Planck's assumption was valid for all electromagnetic waves, not just standing electromagnetic waves. According to Einstein's hypothesis, light is composed of quanta with energy $h\nu$. The number n indicates how many photons the beam contains and this number determines the intensity of the light beam. In this case, the energy of a single photon is determined only by its frequency. With this assumption, it is possible to explain the results of Lenard's experiment.

The American experimental physicist R.A. Millikan opposed Einstein's hypothesis because it contradicted the classical electromagnetic wave theory of light, but after 10 years of experimental work, he obtained results that confirmed Einstein's hypothesis, contrary to his initial expectations. Based on Einstein's assumption, Millikan succeeded in measuring Planck's constant with high precision. Millikan's experimental work on the photoelectric effect is one of the important studies that proved Einstein's hypothesis. These studies were considered by the Nobel committee as sufficient evidence to confirm Einstein's hypothesis about the photoelectric effect and Einstein was awarded the Nobel Prize in physics in 1921. Millikan also received the Nobel Prize in physics in 1923 for his experimental work on the photoelectric phenomenon and electric charge.

Einstein's hypothesis indeed successfully explains the experimental results obtained by Lenard and Millikan on the photoelectric phenomenon. Taking the energy of a photon as $h\nu$, the absorption of a photon by a metal surface increases the energy of an electron in the metal by $h\nu$. Electrons with increased energy cannot leave the metal surface immediately. Because there is a potential energy binding the electrons to the metal surface, it is necessary to do a work of Φ to separate the electron from the metal surface. Φ is the minimum energy required to detach an electron from a given metal surface. This energy is called the “work function”. When the

energy of the electron increases by $h\nu$, Φ of this energy must be spent to detach the electron from the metal.

Work function,

$$\Phi = h\nu_0 \qquad \textbf{(3)}$$

is given as ν_0 is the threshold frequency.

If $h\nu < \Phi$, there will be no electron removal, but if $h\nu > \Phi$, there will be removal, and the remaining energy $h\nu - \Phi$ will be the electron's kinetic energy. In this case the kinetic energy of the photoelectrons,

$$\frac{1}{2}m\mathrm{v}^2 = h\nu - \Phi \qquad \textbf{(4)}$$

can be written. As can be seen, the kinetic energy of the photoelectron is only linearly dependent on the frequency of the light. Photons with a frequency lower than this threshold frequency cannot remove electrons from the metal and the photoelectric effect does not occur. When the intensity of the light beam increases, it is only the number of photons contained in the beam that increases. There is no change in the energy of each photon. In this case, more photoelectrons will be removed from the metal surface, but the kinetic energy of these photoelectrons will not change.

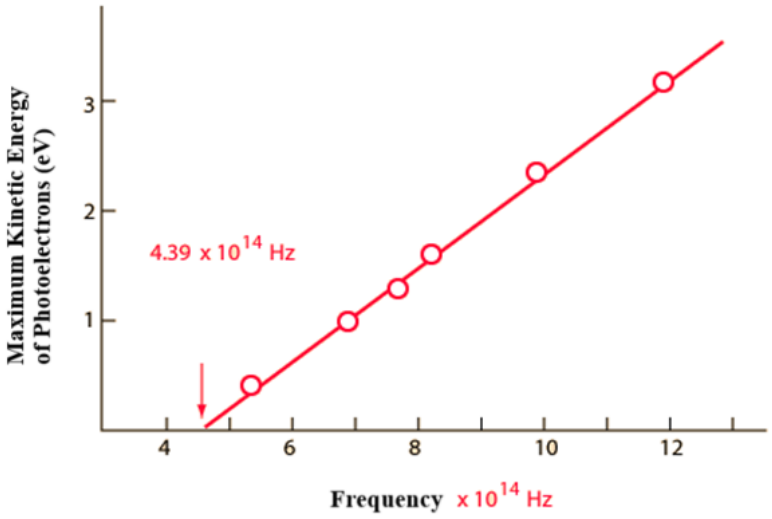


Figure 2: Frequency kinetic energy graph

It can be seen from relation (4) that the relationship between the kinetic energy of photoelectrons and the frequency of light is linear. If the kinetic energy of the photoelectron is plotted against the frequency of the photon, the graph is a linear line, the slope of the graph is Planck's constant and the point where the graph crosses the frequency axis gives the threshold frequency. Figure 2 shows the kinetic energy-frequency plot based on data obtained by Millikan in 1916.

$$eV_d = K_{max} = h\nu - \Phi \qquad \textbf{(5)}$$

From the slope of the graph, we find Planck's constant, $h = 4{,}16 \times 10^{-15}\, eV.s$. This value agrees with the value found by Planck with blackbody radiation. ($h = 4{,}1356675 \times 10^{-15}\, eV.s$ or $h = 6{,}626 \times 10^{-34}\, J.s$)

The findings of Hertz and subsequent experiments to explain the photoelectric phenomenon:

1. The photoelectric phenomenon is a physical phenomenon that supports the particle structure of light.

2. Polished metal surfaces illuminated with visible light and/or UV light emit electrons. These surfaces do not emit positive ions.

3. For these surfaces to emit electrons, a threshold energy (or a corresponding threshold frequency) must be exceeded.

4. The kinetic energy of the electrons (photoelectrons) detached from the surface does not depend on the light's intensity but on the light's frequency.

5. In the experiments, the time difference between the light hitting the metal surface and the electron detachment is ~10^{-9} s. This time is independent of the frequency of the light.

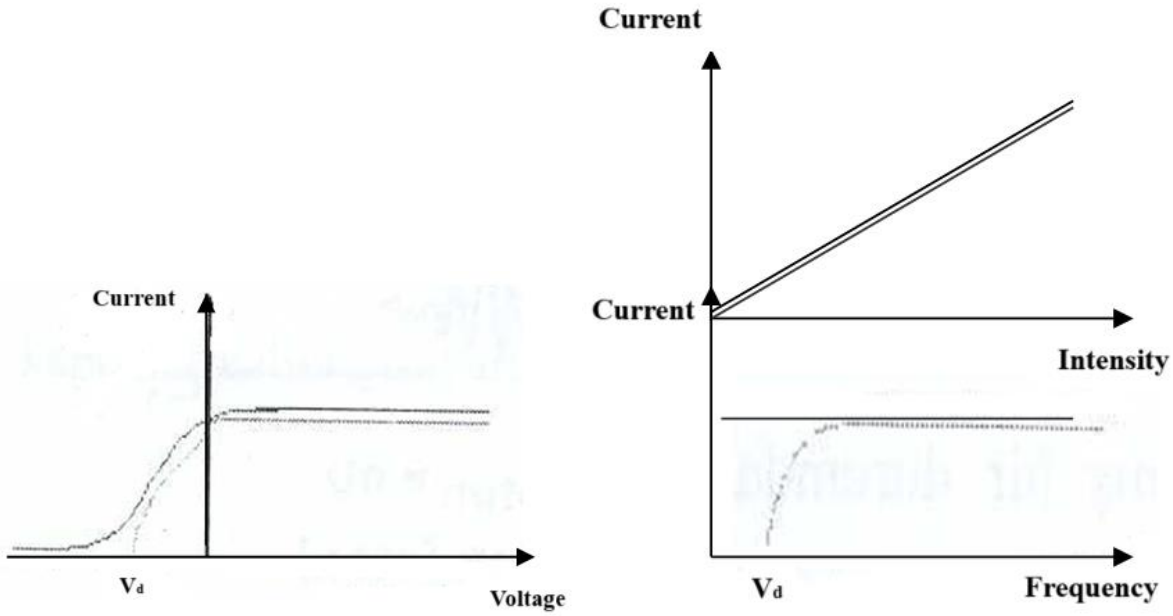


Figure 3: Variation of current with voltage, light intensity, and frequency of light

The current generated by photoelectrons depends on the light's intensity if the light's frequency is greater than a threshold value. As the intensity of the light increases, the current increases. In classical physics, the removal of electrons from a metal surface must occur at any frequency depending on the light intensity, whereas in the photoelectric event, electrons can be removed above a certain threshold value.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup to be used is given in Figure 4.



Figure 4: Experimental set-up for determining Planck's constant (h) by photoelectric effect.

Elements in the optical setup shown in Figure 4:

1. Power Supply, **2.** Mercury Lamp (80 W), **3.** Slit (Adjustable), **4.** Convex Lens (f= +100 mm), **5.** Grating (600 lines/mm), **6.** Photocell, Electrometer (R>1013 Ω), **7.** Digital Multimeter, **8.** Filter (525 nm and 580 nm)

Steps

1. Turn on the Mercury Lamp and wait approximately 15 minutes for it to reach full brightness. If this lamp is turned off for any reason, it will need to cool down before it can light up again. Therefore, the lamp should not be turned off during the experiment.

2. Before starting the measurements, arrange the optical setup so that it is in the position shown on the goniometer.

3. A diffraction pattern appears on the screen. Rotate one arm of the optical assembly so that the entrance of the photocell overlaps the visible region spectral lines of the 1st order in the diffraction pattern on the screen. Read the angles at which each spectral line is observed with the help of the goniometer on the experimental setup and note them in Table 1.

$$m\lambda = d\sin\theta \qquad \textbf{(6)}$$

Using Bragg's law (relation (6)), calculate the wavelength and frequency values corresponding to each angle value and note them in Table 1. Check that each calculated wavelength corresponds to the spectral line spacing in Figure 5.

4. Move the optical setup over each spectral line and read the corresponding constant voltage values with a multimeter and note them in Table 1. (Wait for a few seconds to obtain constant and accurate voltage values).

5. To avoid 2nd-order UV diffraction phenomena, measurement values for the yellow and red spectral lines are obtained with the help of filters. Place a 525 nm wavelength filter for the yellow spectral line and a 580 nm wavelength filter for the red line at the inlet of the photocell read the corresponding constant voltage value and note it in Table 1.

6. Using the values obtained, form equation (5) for each color given in Table 1. Calculate 4 Planck constants by solving the possible binary equation systems obtained.

Example binary equation system:

$$eV_{d(violet)} = h\nu_{violet} - \Phi \qquad \textbf{(7)}$$

$$eV_{d(yellow)} = h\nu_{yellow} - \Phi$$

7. Draw the graph between the obtained stopping potential values and the frequency of light. Using the slope value in the graph, calculate Planck's constant and show that Planck's constant is compatible with the real value.

8. Calculate the metal's work function and threshold frequency in the photocathode.

Table 1: Results

Color	**Angle**	**Wavelength λ(nm)**	**Frequency ($\times 10^{14}$ Hz)**	**Stopping Potential (V)**
Violet				
Green				
Yellow				
Red				

$h_{experiment} = h \mp \Delta h$ J.s $\qquad$ $h_{graphic} = \ldots\ldots\ldots$ J.s

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